

E-Commerce Sales Performance Analysis & Statistical Validation

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Agenda

01

Dataset Overview

Structure, features & data types explored

02

Gender Distribution

Female 514 | Male 486 — near-equal split

03

Sales by Category

Electronics dominates with ₹3.7M in sales

04

Sales by City

Hyderabad leads; Chennai trails the field

05

Customer Age Distribution

Age 60 segment is the highest customer group

06

Monthly Sales Trend

Peaks in May & July; dip in Oct 2025

07

Gender Spending Comparison

Box plot analysis of purchase amounts

08

Hypothesis Testing

T-Test results & statistical conclusions

Dataset Overview

Dataset Columns (Index)

Order_ID

Customer_ID

Gender

Age

City

Category

Purchase_Amount

Order_Date

1,000

Total Orders

514 / 486

Female / Male

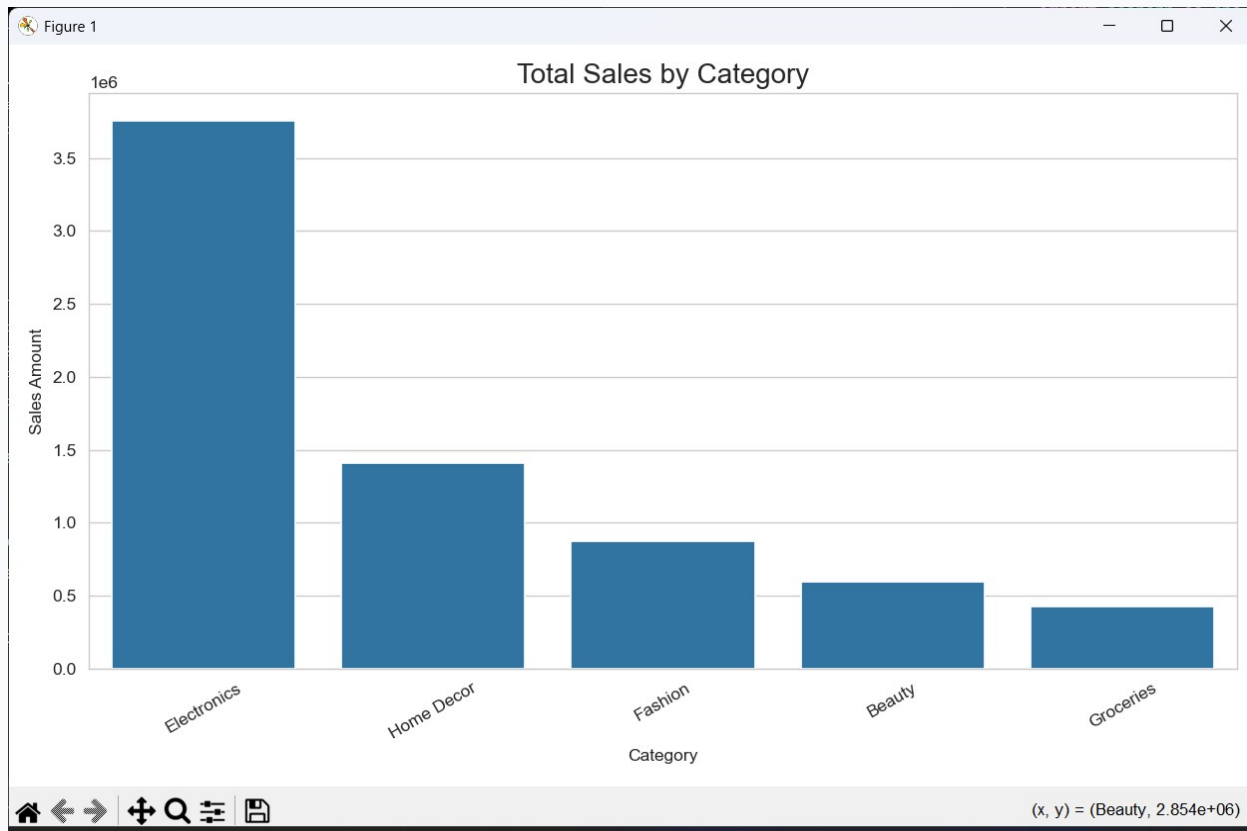
8 Cities

Coverage

5 Categories

Product Types

Total Sales by Category



Electronics

₹3.7M — #1 category

Home Decor

₹1.4M — 2nd place

Fashion

₹0.87M

Beauty

₹0.59M

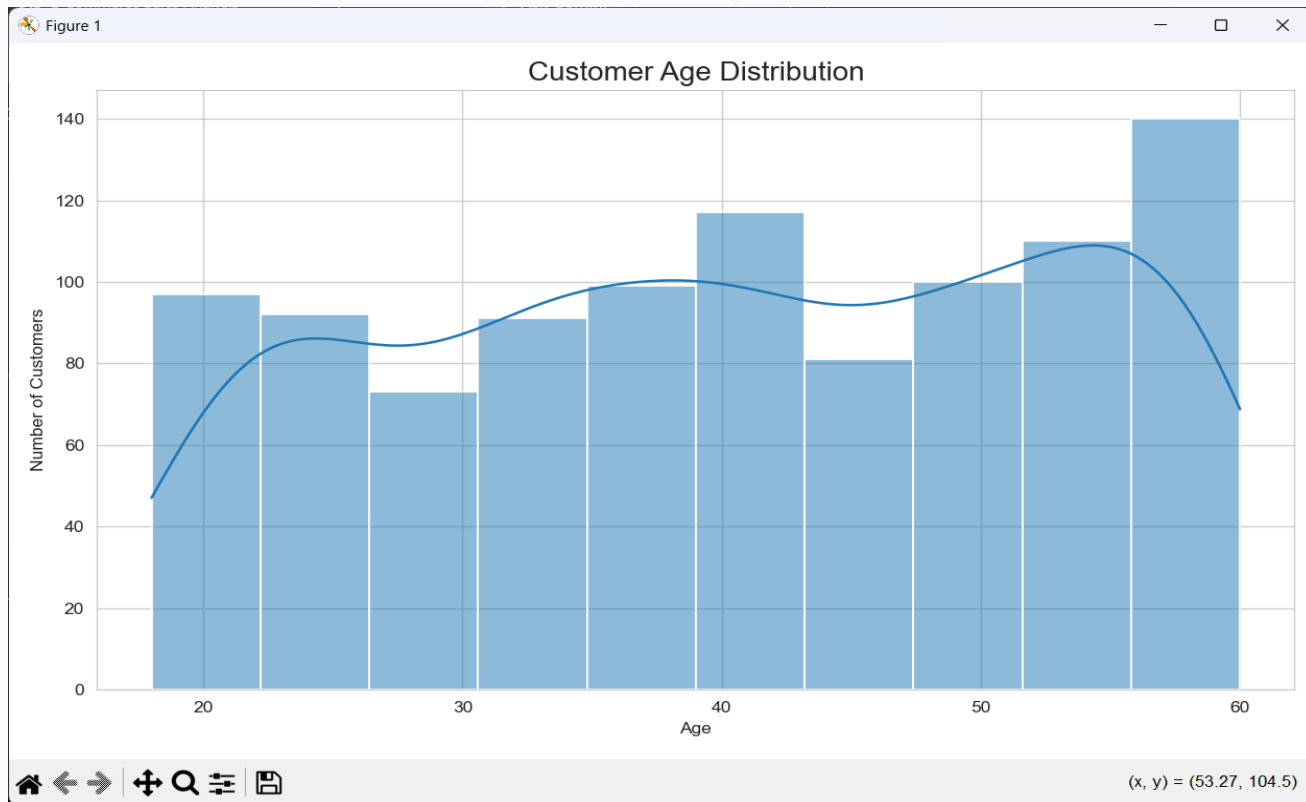
Groceries

₹0.43M

Total Sales by City



Customer Age Distribution



Age Group Analysis

60+

Highest segment — 140 customers

40s

Notable spike — ~117 customers

50s

~110 customers

20s-30s

~73-97 customers

Monthly Sales Trend (2025)

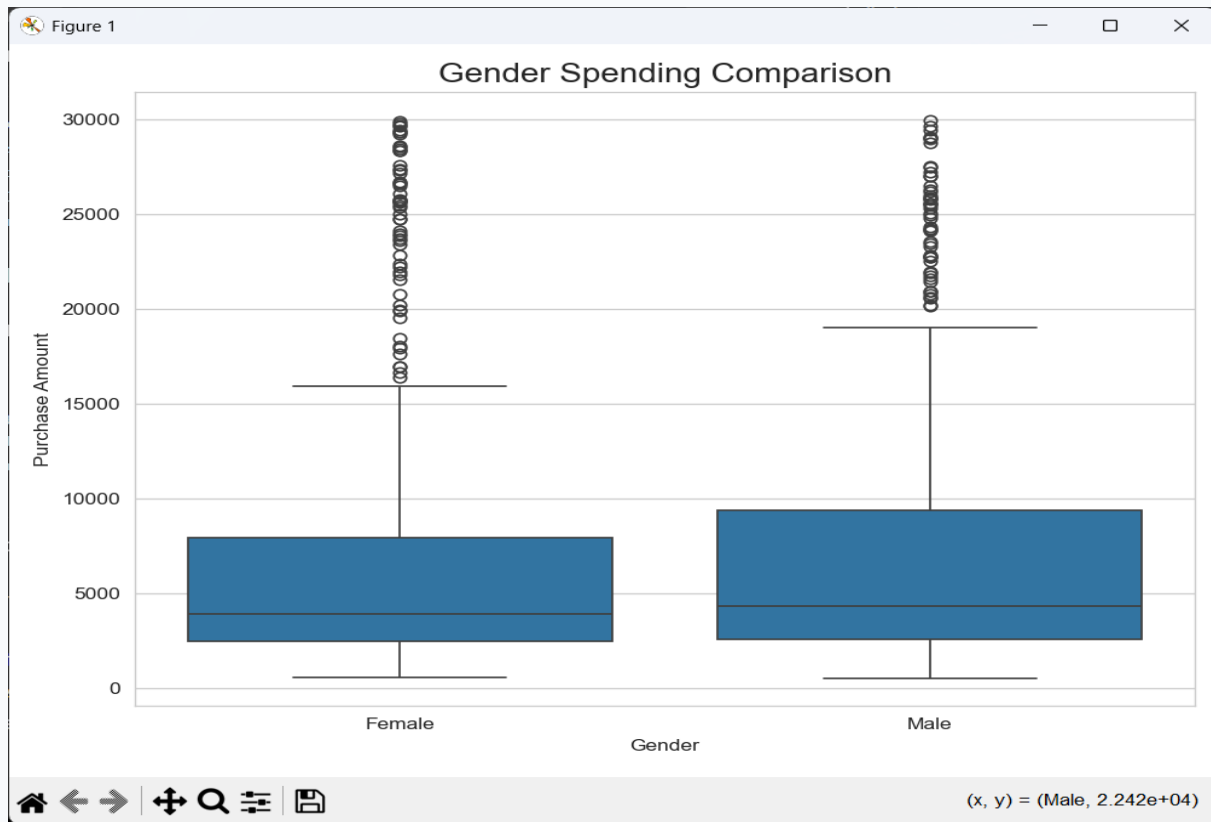


▲ **Peak Months:** May (₹695K) & July (₹677K) — highest performance

▼ **Dip Period:** Oct 2025 (₹462K) — sharpest drop in the year

📈 **Recovery:** November rebounds to ₹612K after the Oct low

Gender Spending Comparison



Box Plot Summary

Female

Median: ~₹4,000
IQR: ₹750–₹8,000
Whisker up to ₹16K

Male

Median: ~₹4,500
IQR: ₹700–₹9,500
Whisker up to ₹19K

Both genders show many high-value outliers above ₹20K

Hypothesis Testing Results

H_0 (Null Hypothesis)

There is NO significant difference in spending between Male and Female customers.

H_1 (Alternate Hypothesis)

There IS a significant difference in spending between Male and Female customers.

Independent Samples T-Test Results

Metric	Male Samples	Female Samples	T-Statistic	P-Value
Value	486	514	1.2589	0.2084

✓ Conclusion

Since p-value (0.2084) > α (0.05), we **FAIL TO REJECT the Null Hypothesis (H_0)**. There is no statistically significant difference in purchase amounts between Male and Female customers.

Both genders spend similarly on average, despite the slight IQR difference visible in the box plot.

Key Takeaways



Electronics dominates sales at ₹3.7M — 3× higher than any other category



Hyderabad is the top-performing city; geographic diversification needed



Age 60+ is the largest customer segment — target for premium products



May & July peak; aggressive promotions needed for Oct-Sep trough



No significant gender-based spending difference ($p = 0.2084 > 0.05$)

Thank You!